

Molecular QTL mapping in humans

Odysseas Sotirios Stergiou

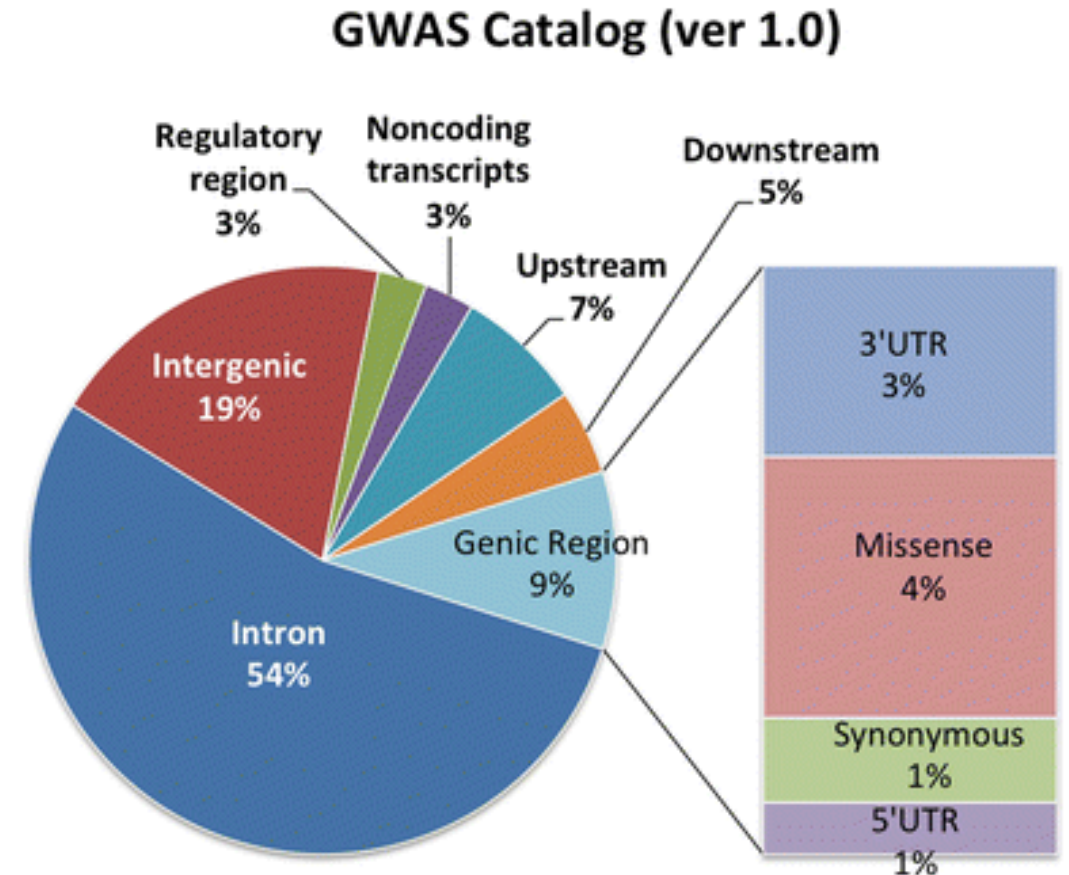
VSS 2026

Agenda

1. **Core Concepts & Definitions** → molQTL, eQTLs vs pQTLs, cis vs trans
2. **Data Generation & QC** → RNA-seq normalisation, genotype arrays/WGS, covariates, PCA
3. **Association Statistics** → Linear regression framework, millions of tests, permutations & FDR
4. **Validation & Biological Checks** → Distance-to-TSS, enrichment in regulatory elements
5. **From QTLs to Trait Biology** → eQTL – GWAS SNP colocalization , GTEx

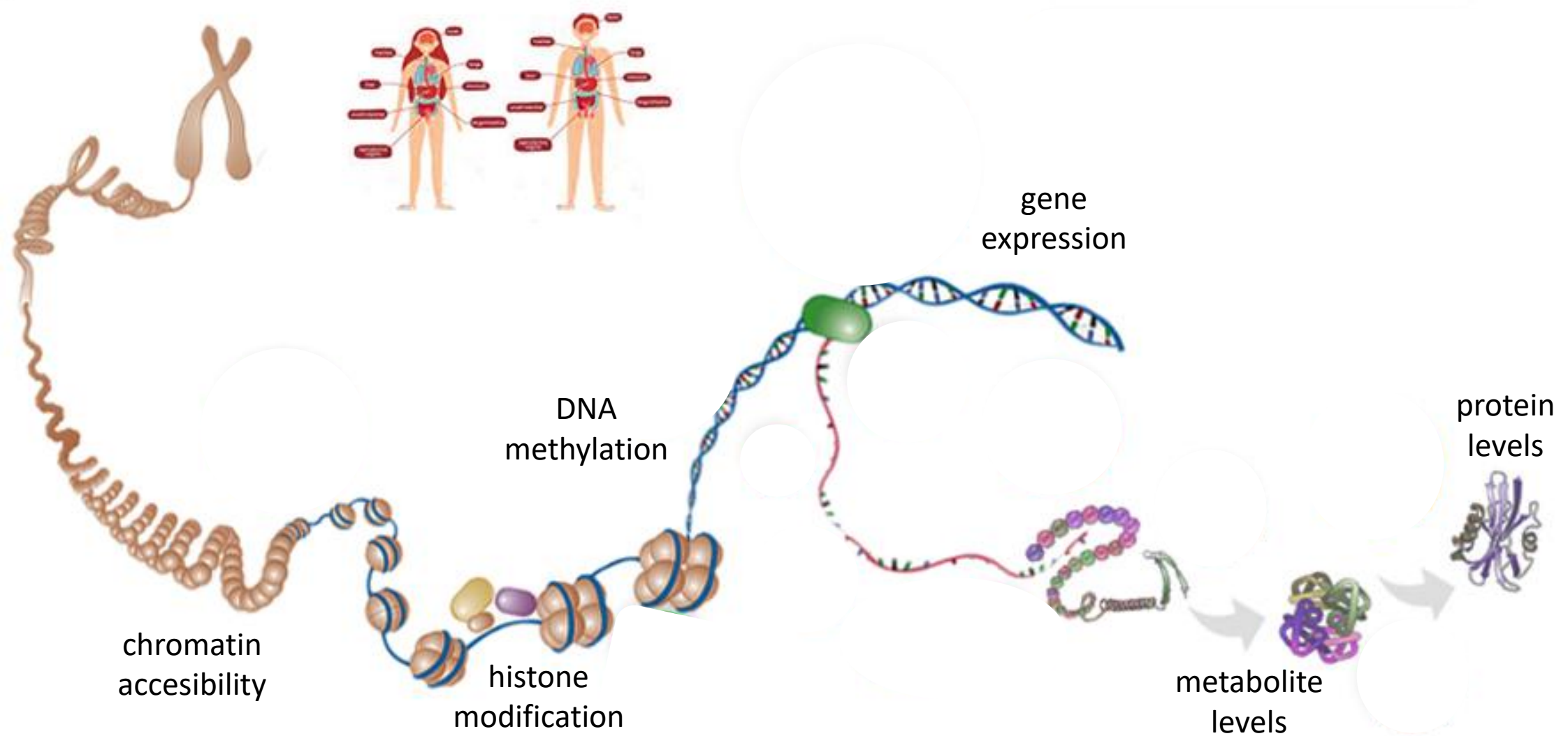
How do genetic variants affect cellular processes?

- Few GWAS variants shape phenotypes in a straightforward way (protein-changing variants).
- In most cases, a GWAS is performed on a complex disease, i.e. several (hundreds) SNPs contribute to the phenotype.
- Most of disease-associated (GWAS-detected) variation is in non-coding regions
- To understand functional effects of such genetic variants → molecular quantitative trait locus (molQTL) mapping



GWAS tells us **where** genetic signals are.
Molecular QTL mapping tells us **how** they might be acting.

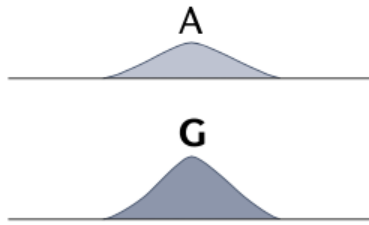
Multiple molecular traits in cells can be quantified



Quantifying molecular traits – can be measured at scale

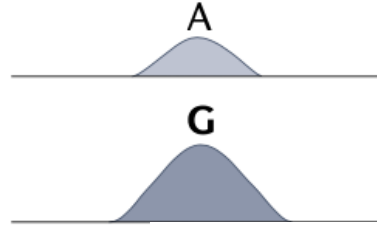
Chromatin accessibility QTL (caQTL or chQTL)

Chromatin accessibility measured by ATAC-seq, DNase I sensitivity, etc.



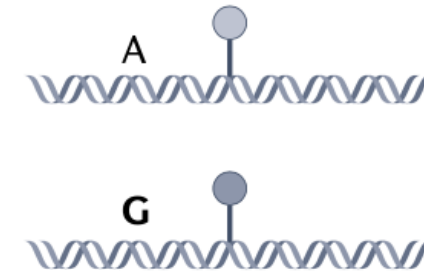
Histone modification QTL (hQTL or cQTL)

Histone mark ChIP-seq peak height



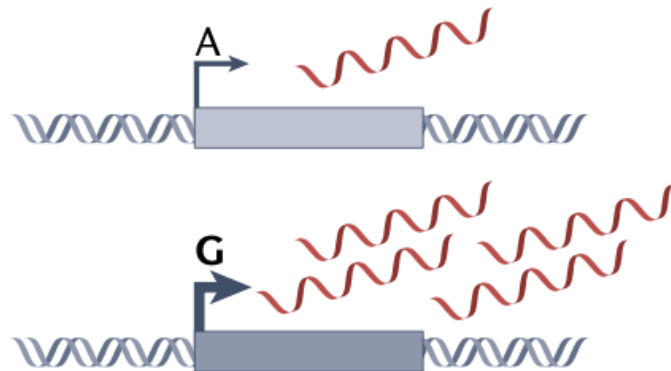
Methylation QTL (meQTL)

Methylation ratio of a CpG site



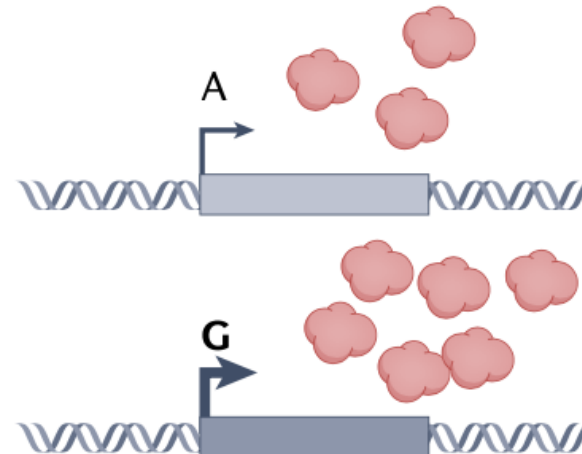
Expression QTL (eQTL)

RNA expression level of a gene or a transcript



Protein QTL (pQTL)/metabolite QTL (mQTL)

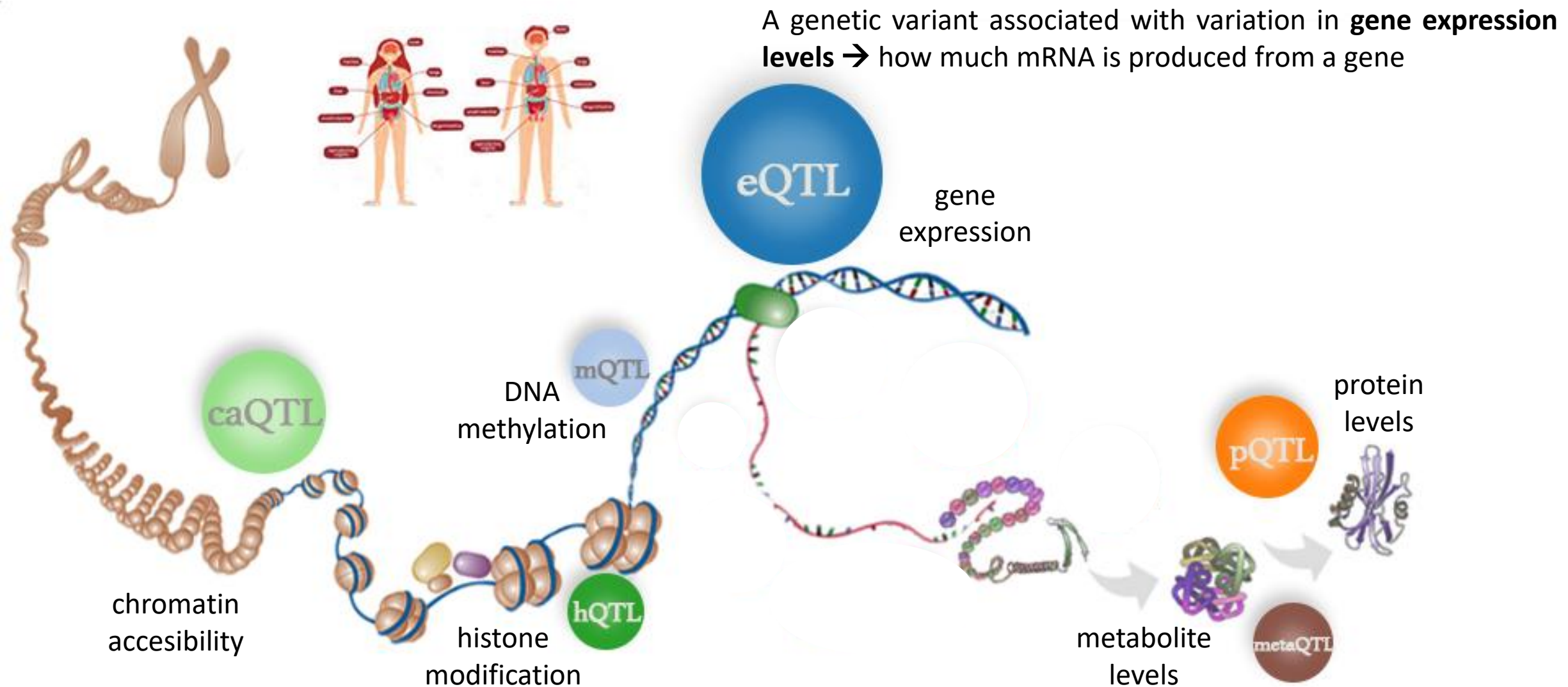
Protein expression level of a gene



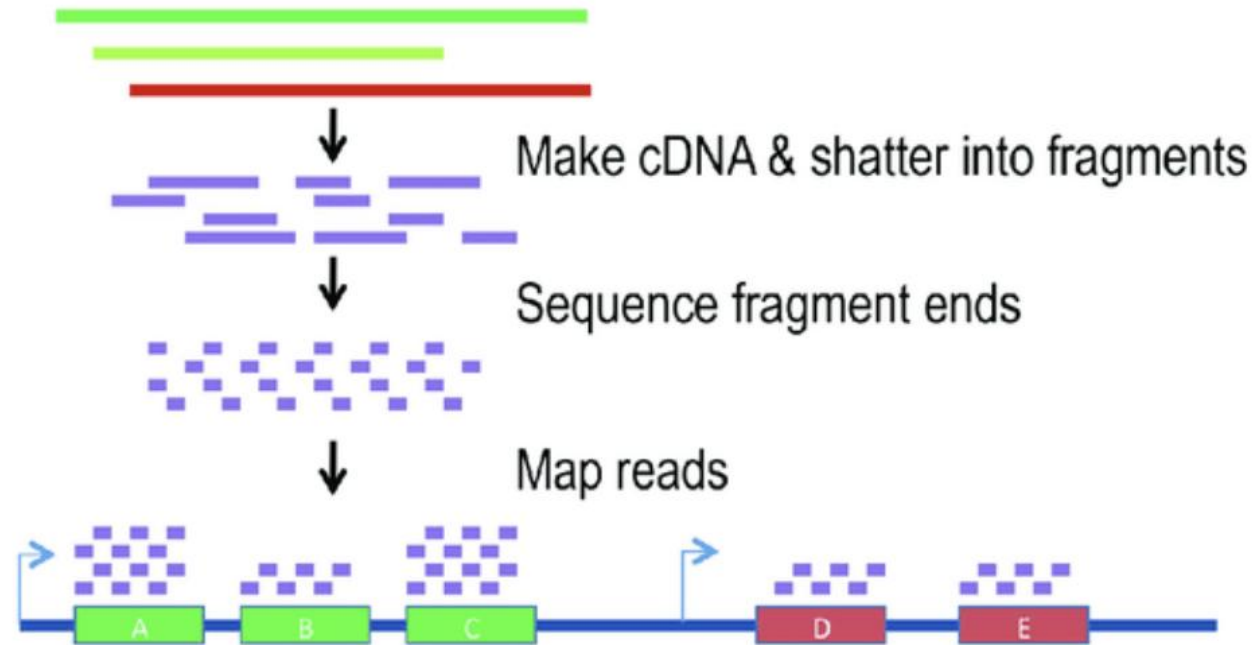
Genetic associations for molecular traits can be mapped in a similar way to GWAS

- Molecular traits have substantial genetic component → map genetic associations in **similar way to GWAS**
- Statistically, GWAS for a quantitative trait (e.g. height) and QTL mapping are nearly identical approaches
→ **Linear regression to associate genetic variation with a quantitative phenotype** in a population sample
- Typically:
 - **GWAS** used for *non-molecular* traits (e.g. height)
 - **QTL** refers to *molecular* quantitative trait locus
- molQTL umbrella term for loci with a genetic association for a quantitative level of a molecular trait including:
 - eQTLs – gene expression
 - mQTLs – DNA methylation
 - hQTL – histone modification
 - caQTLs – chromatin accessibility
 - pQTLs – protein levels
 - metaQTLs – metabolite levels

molQTL umbrella term for loci with a genetic association for a quantitative level of a molecular trait

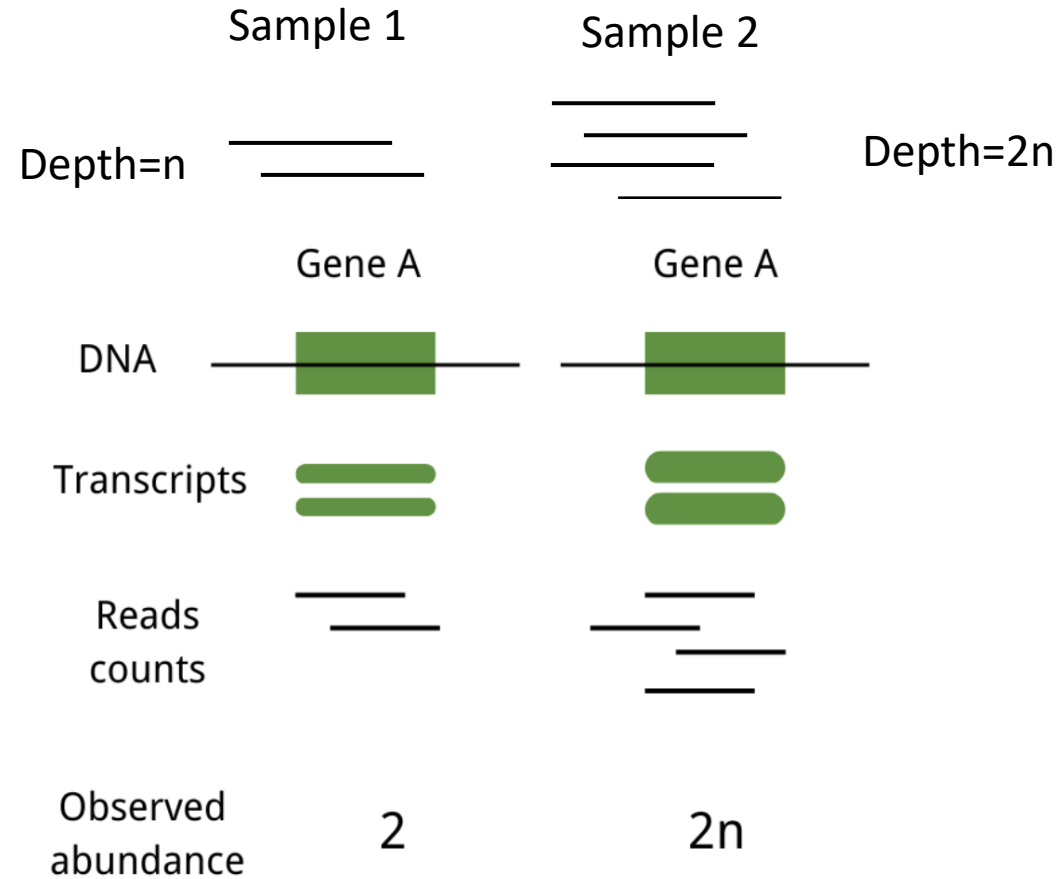


Measure gene expression through RNA-Seq in specific tissue or cell type



RNA quantification: prepare count matrix with expression levels for each gene in each individual

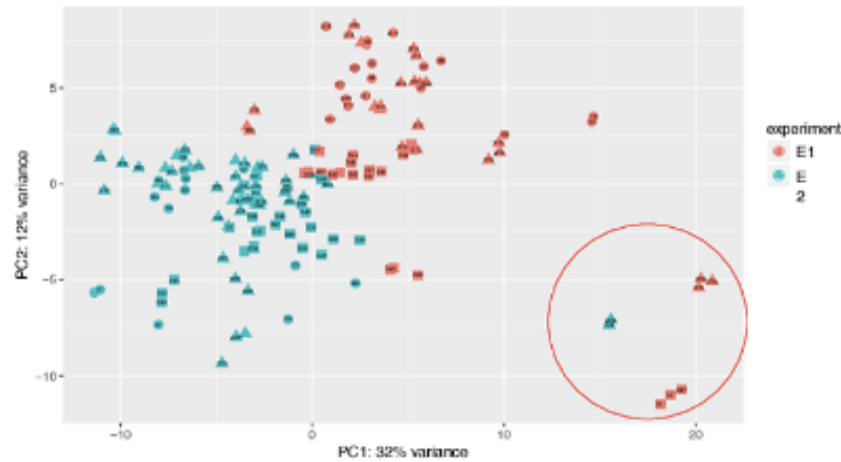
Sequencing depth and abundance



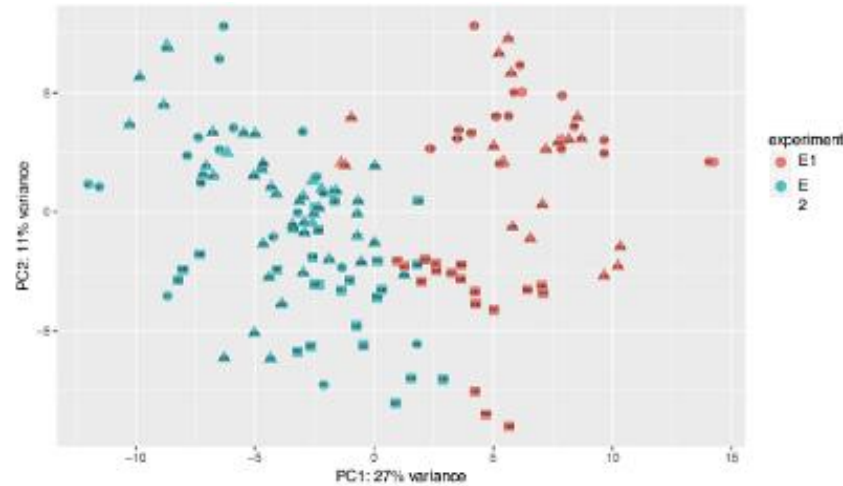
- As overall sequencing depth increases, Gene A sequencing depth increases, more read counts are produced
- Need to normalise for sequencing depth (gene counts are scaled by scaling factor), samples comparable
- Inverse normal transformation of gene expression to conform to assumptions of regression model (homoscedasticity)

Between sample normalization

- QC to exclude problematic samples (may result in loss of power or introduction of artefacts)
- Samples mostly fail due to problems with original biospecimen (e.g. RNA degradation)
- Inspect data for outliers and batch effects using PCA



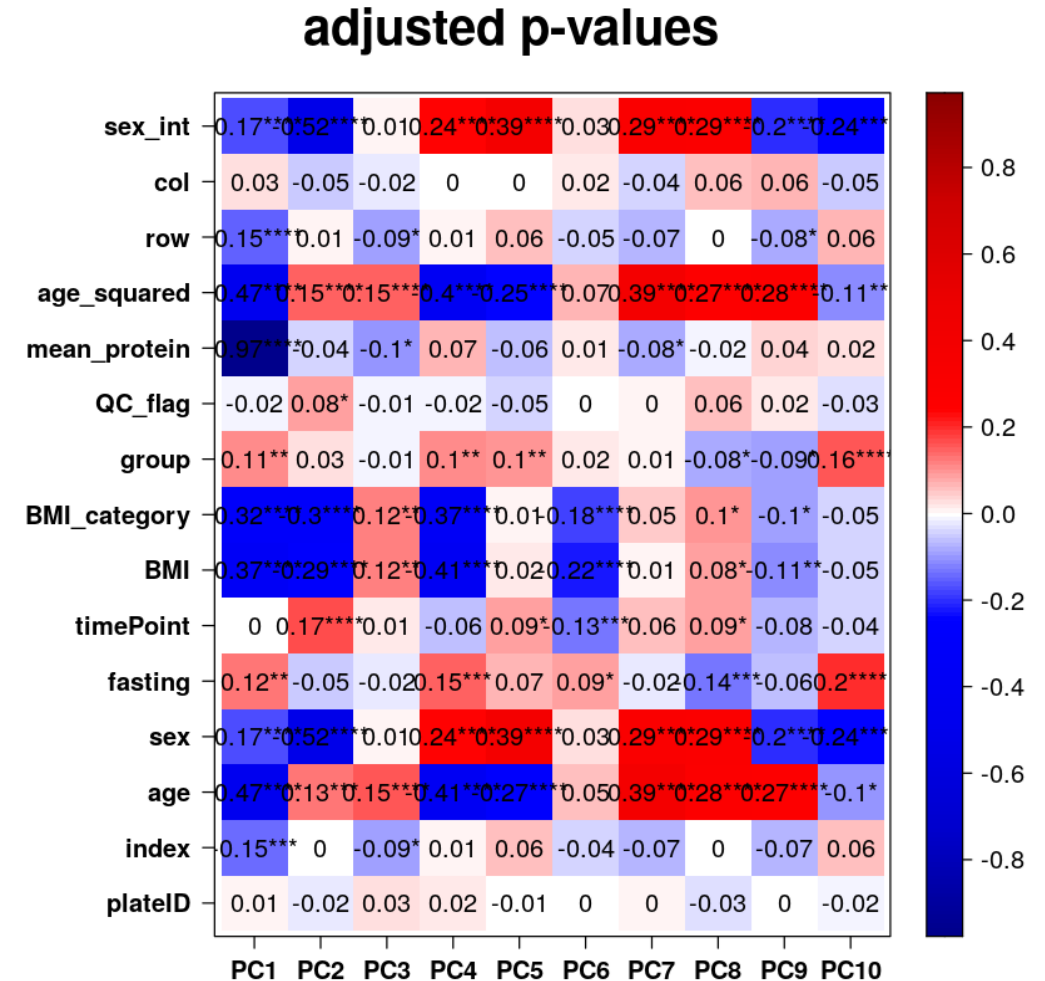
with outliers



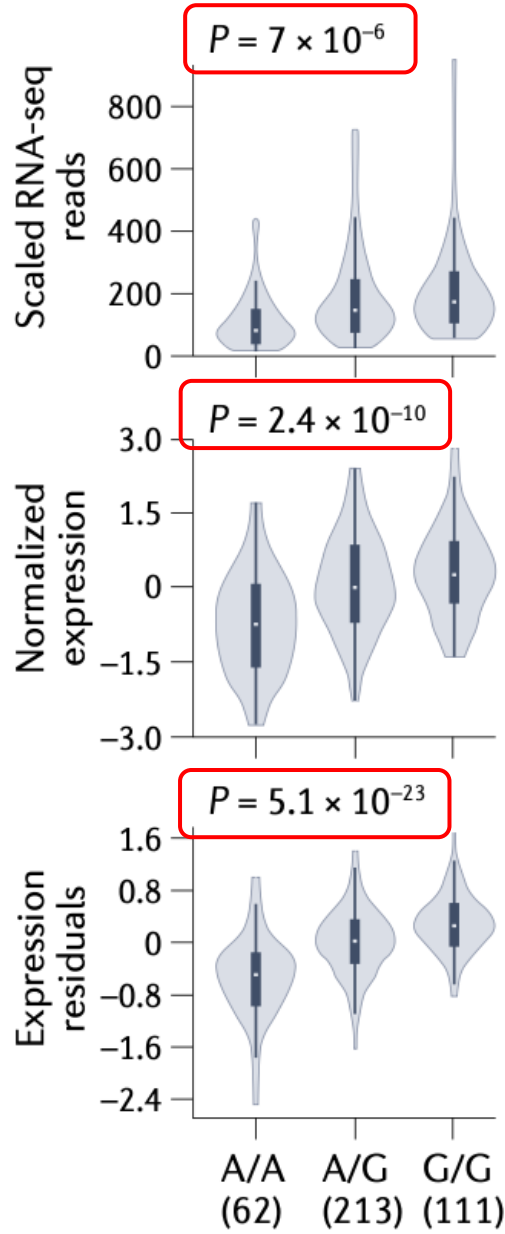
without outliers

Correcting for confounding factors (covariate inclusion)

- Confounding effects can lead to loss of power, false positive associations
- Include covariates in association testing that account for effects of confounders
- Known confounding effects of e.g. **age** and **sex** on gene expression
- Better to correct using latent variables computed from the normalized expression data using:
 - principal components (PCs)
 - probabilistic estimation of expression residuals (PEER) factors



Gene expression as function of genotype – effect of QC, normalization and correction for covariates

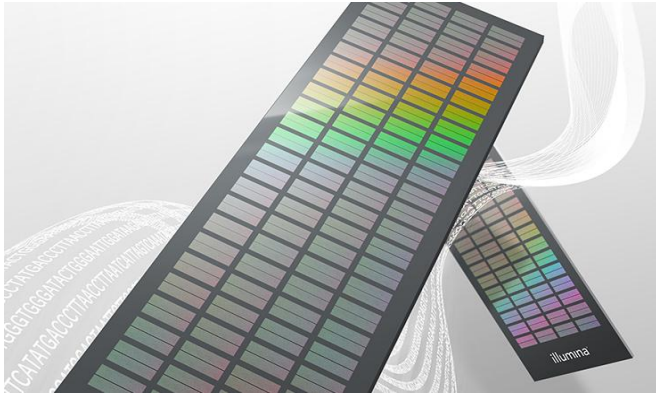


Association test on gene expression (RNA-Seq) corrected only for library size

Association test on inverse normal transformed gene expression values

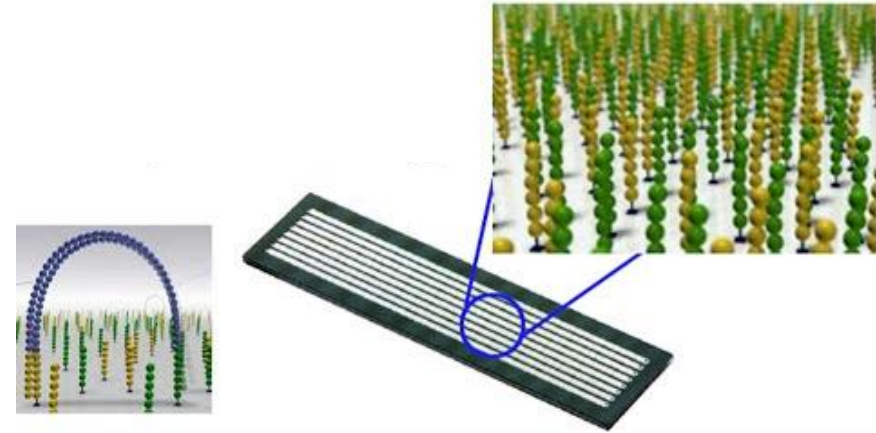
Association test on gene expression residuals after covariate correction

Identify genetic variants through SNP arrays or whole genome sequencing



SNP arrays

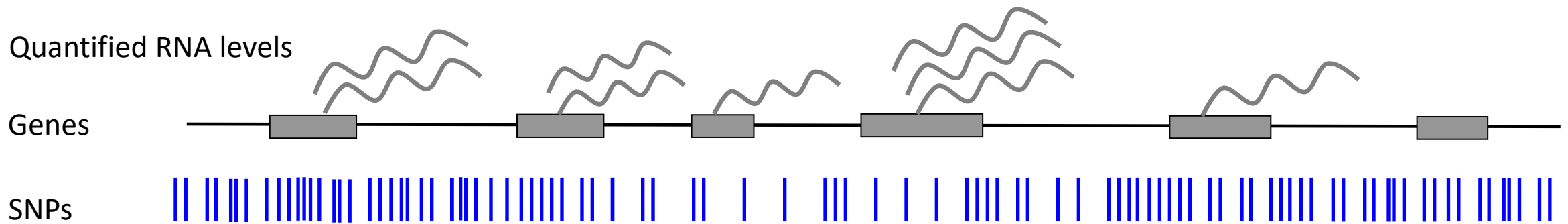
- Target hundreds of thousands to few million known genetic variants
- Genotype + imputation to obtain good coverage of common variants



Whole genome sequencing

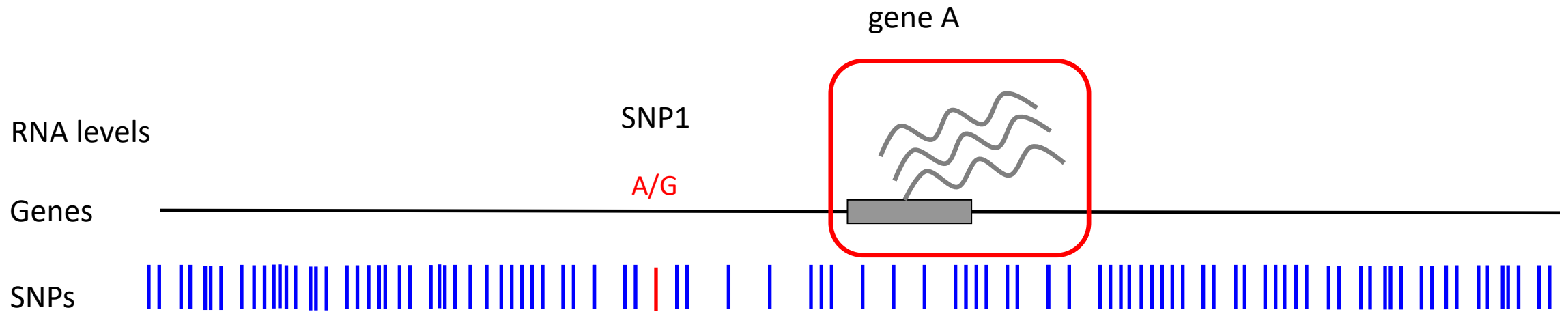
- Provides increased power for identifying causal variants
- Provides possibility to map QTL effects for complex genetic variants (including short tandem repeats, indels, structural variants)

Genome-wide association of genetic variation with levels of gene expression (molecular trait) to map eQTLs (molQTLs)



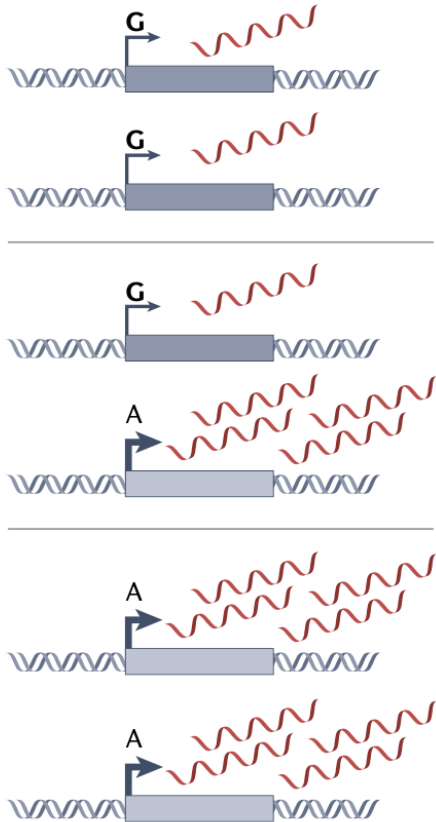
- Association of millions of SNP with tens or hundreds of thousands of molecular features across the genome (all expressed genes in a tissue) through linear regression
- Typically, minimum sample size of ~70 samples to detect strong eQTLs

Association of SNP genotype to levels of gene expression to map eQTLs

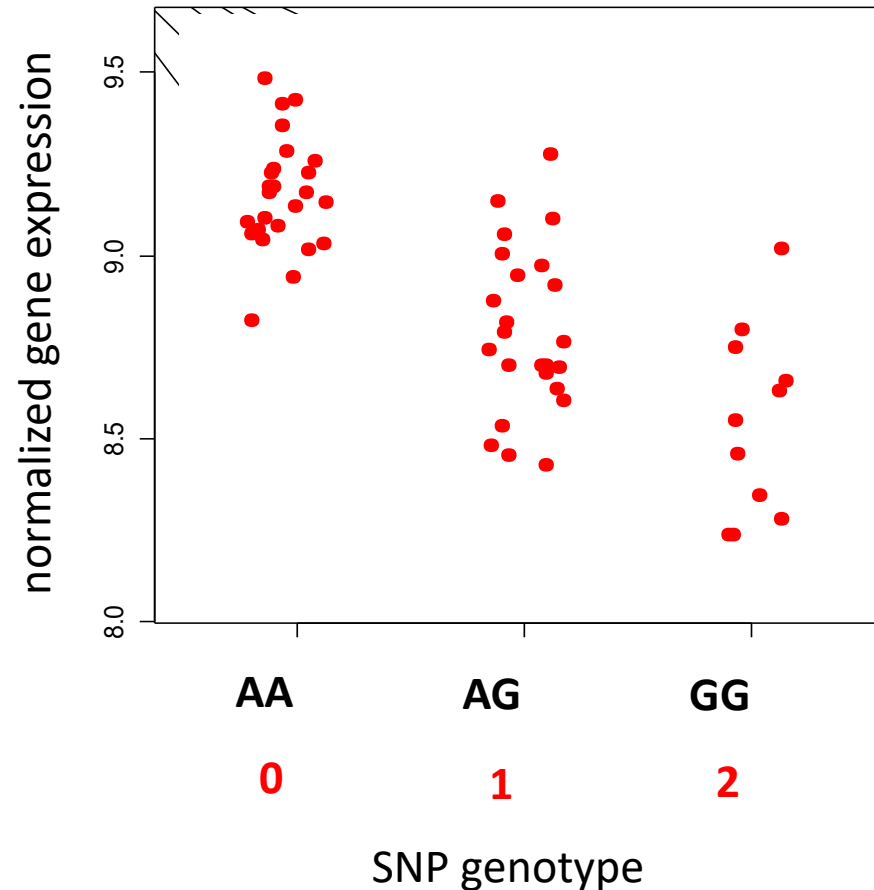


- Test for association between SNP genotype at SNP1 with gene expression at gene A using linear regression

Linear regression to associate SNP genotype with gene expression



Encode genotypes as 0 (AA), 1 (AG), 2 (GG)



Linear regression (LR)

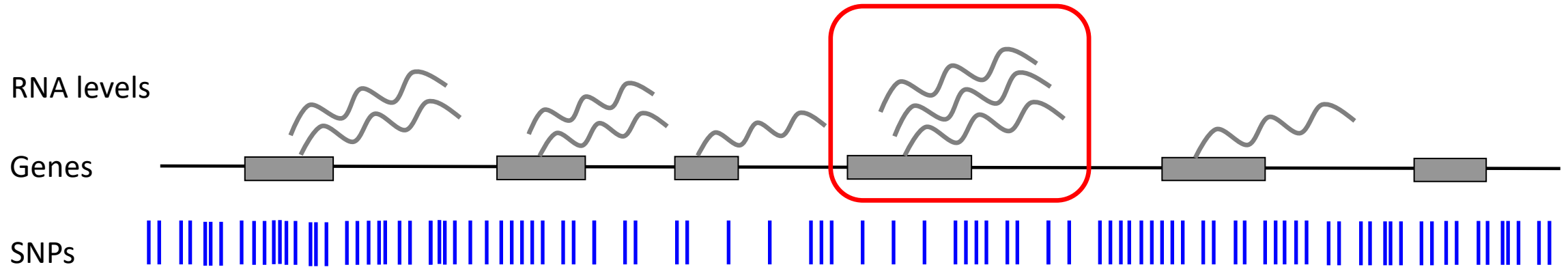
Expression $\sim$ Genotype + Covariates*

- slope
- observed p-value
- r^2 (variance explained)

Assumes additivity of genetic effects

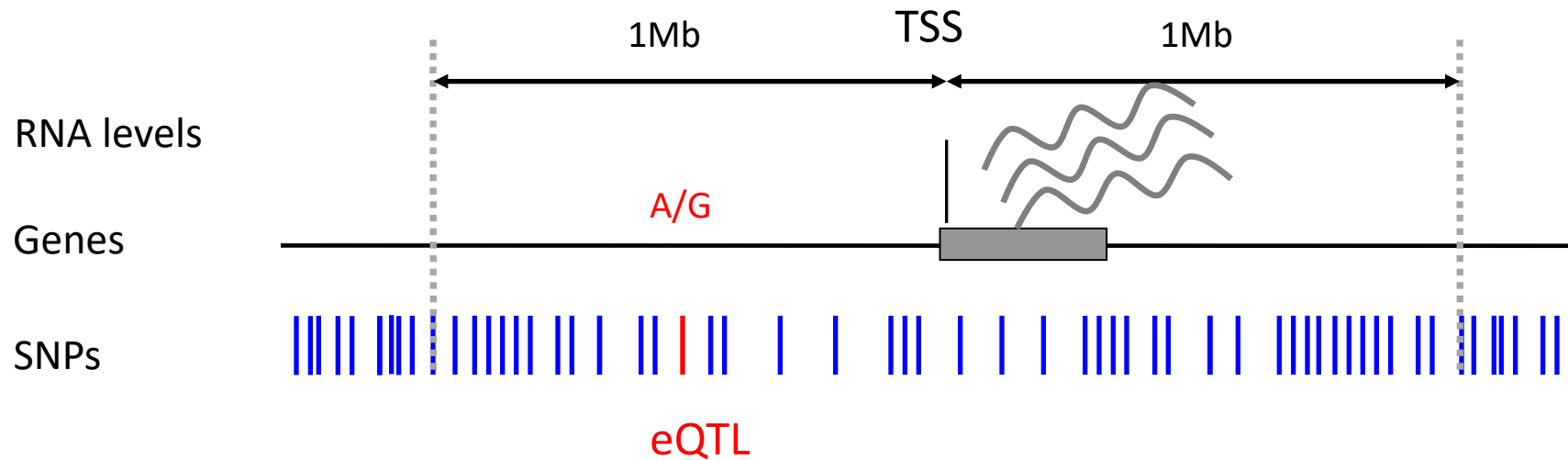
Software: FastQTL, Tensor eQTL, Matrix eQTL

Genes (molecular features) have predefined locations in the genome → **association testing** can be done in *cis* and *trans*



- Typical eQTL study includes:
 - 1×10^7 common variants (MAF $\geq$ 5%)
 - 20,000 expressed genes (molecular phenotypes)

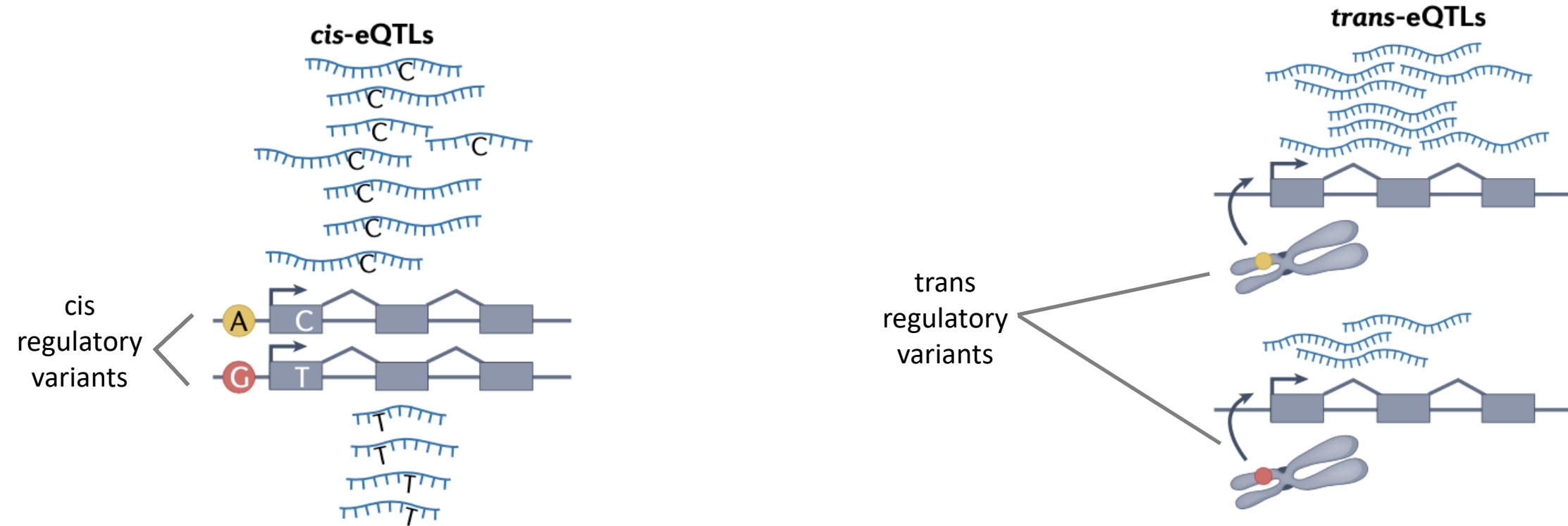
Cis association testing looks for proximal effects to feature studied, ***trans*** looks for distal effects



- ***Cis* testing/mapping** involves testing for association between gene expression (feature) in a 2 Mb window (max) centered on the gene's TSS → $\sim 2 \times 10^8$ (**200 million**) tests for all variant-phenotype pairs in *cis* (assuming 1×10^4 SNPs in each *cis* window)
- ***Trans* testing/mapping** tests for associations between a feature and SNPs located at least 5 Mb away, or on another chromosome
→ 2×10^{11} (**200 billion**) tests for all variant-phenotype pairs in *trans*

Cis variants are typically **close** to their **molecular targets**, *trans* variants are usually on a **different chromosome**

But exact biological definition is not dependent on distance



Alleles A/G of a cis-regulatory variant (eQTL) affect a molecular feature (gene expression) via cis-regulatory activity of the physically connected chromatid

Trans-acting regulatory variants usually act via intermediary molecules

Assigning significance: permutations & FDR

GWAS uses one fixed cutoff ($p < 5 \times 10^{-8}$). We can't do the same, we test hundreds of SNPs per gene, so something looks significant by luck. **Solution:** see how good a result luck alone produces.

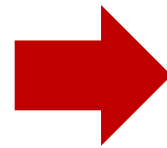
STAGE 1 within one gene → permutation gives one empirical p-value

Person	SNP1 ... SNP10,000	Expression
P1	AA AG GG ...	9.1
P2	AA GG AG ...	9.3
P3	AG AA GG ...	8.4
P4	AG AG AA ...	8.5
...	...	...
P99	GG AG AA ...	8.2
P100	GG GG AG ...	8.1

REAL data

test all 10,000 SNPs → keep smallest: $p_{\text{obs}} = 2 \times 10^{-7}$

genotypes stay fixed



SHUFFLE
expression
column

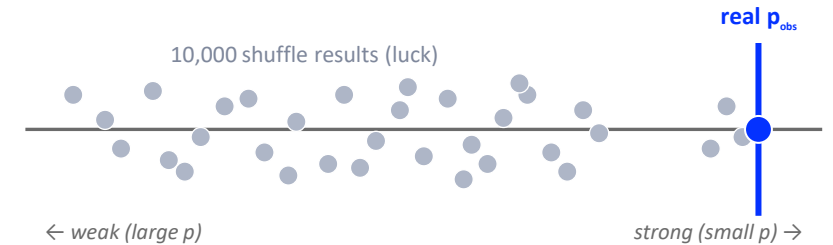
repeat 1,000–10,000×

Person	SNP1 ... SNP10,000	Expression
P1	AA AG GG ...	8.4
P2	AA GG AG ...	8.1
P3	AG AA GG ...	9.3
P4	AG AG AA ...	8.2
...	...	...
P99	GG AG AA ...	9.1
P100	GG GG AG ...	8.5

SHUFFLED (link broken)

test all 10,000 SNPs → keep smallest: 1 luck p

Each shuffle gives one "luck" p-value. Where does the real one land?



only 3 of 10,000 shuffles beat it

empirical $p = 3 \div 10,000 = 0.0003$

STAGE 2 across all ~20,000 genes → FDR controls false hits

One p per gene

Stage 1 gives each gene a single empirical p-value (already corrected for its many SNPs).

Apply FDR < 5%

Benjamini–Hochberg ranks all 20,000 and sets the cutoff from the data, not a fixed line.

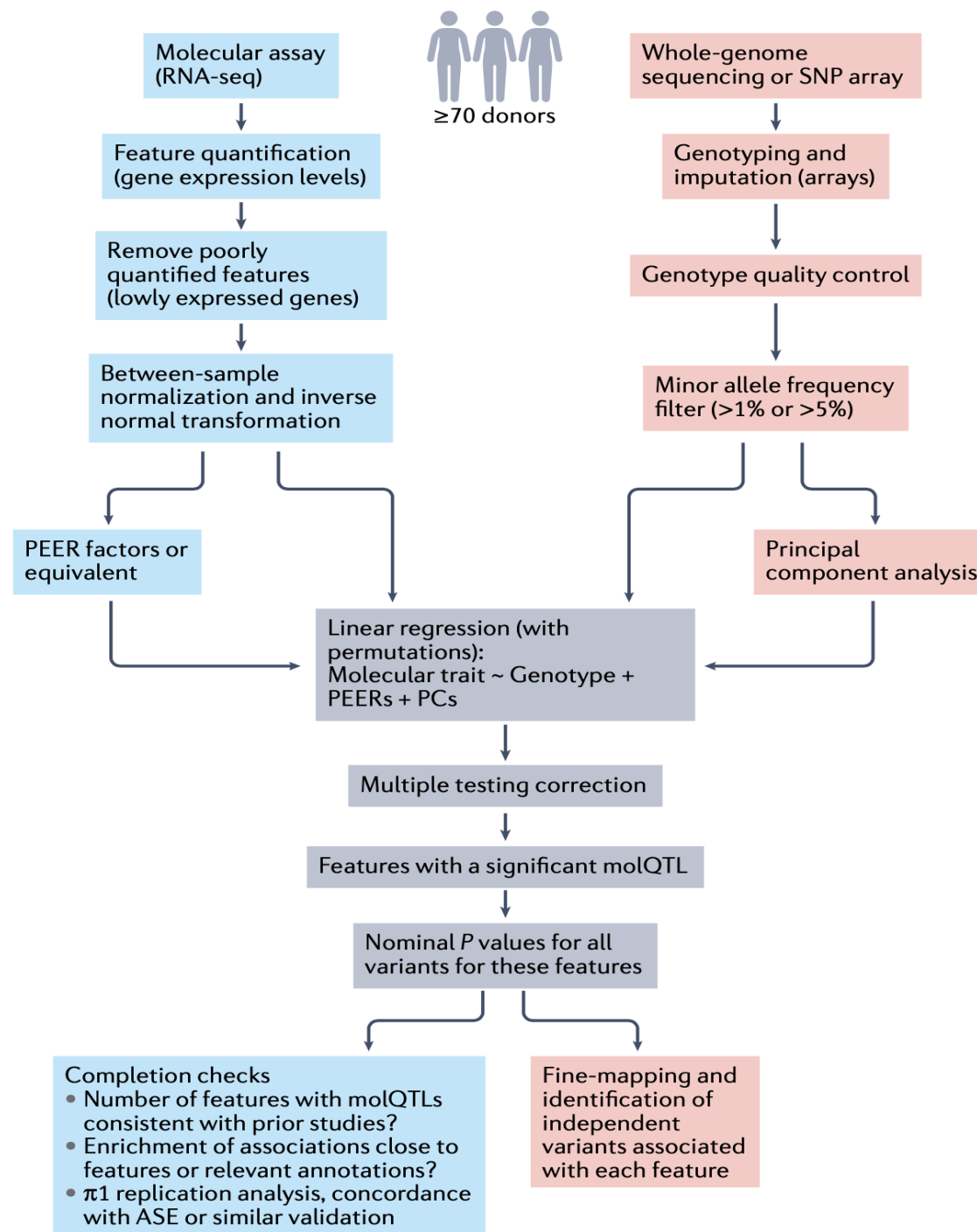
Get eGenes

Genes that pass = eGenes. Meaning: $\leq 5\%$ of the genes we keep are expected to be false.

Speed-up: FastQTL fits a beta distribution to a few thousand shuffles instead of running millions, same answer, far faster.

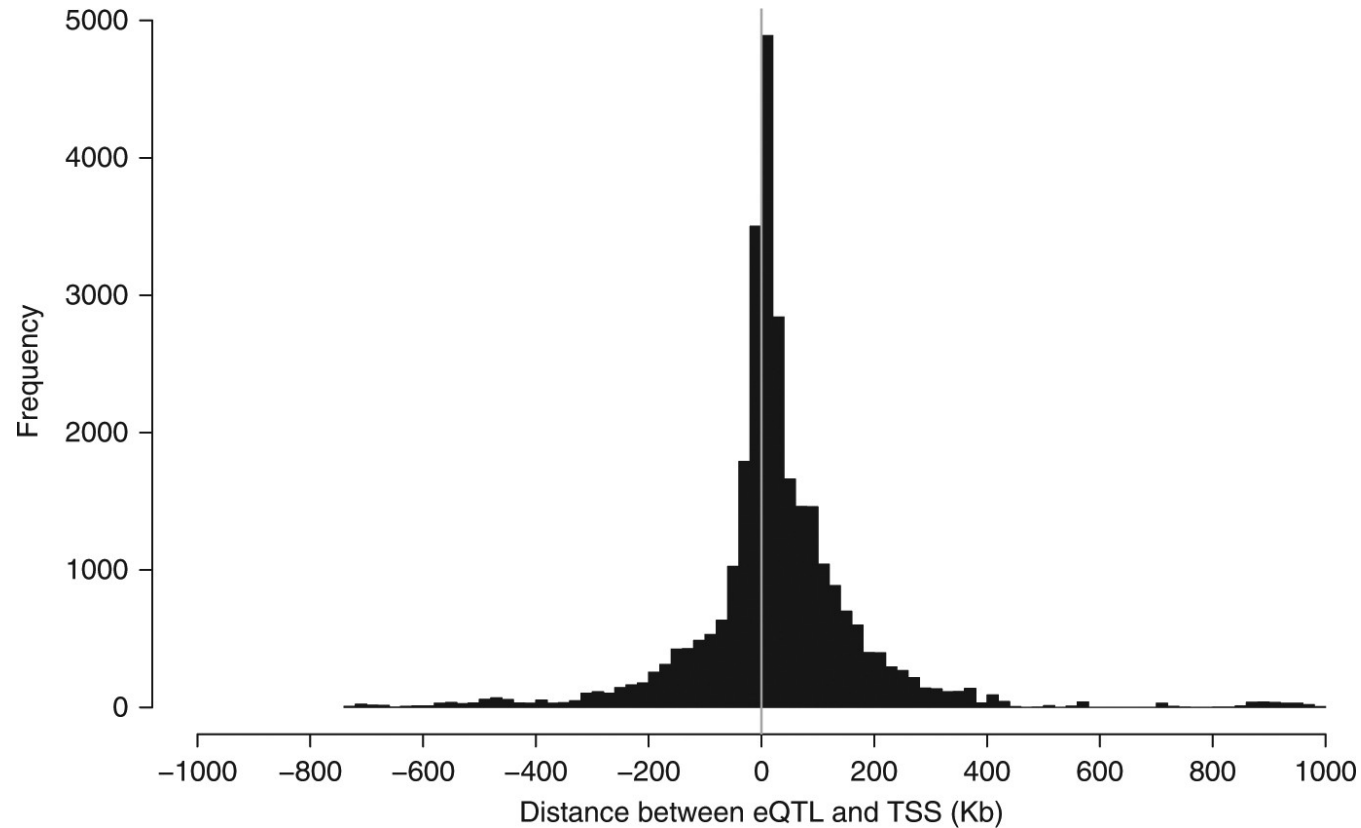
eQTL mapping workflow

- What tissue to study?
- How many individuals?
- Collect molecular trait data
- QC on molecular trait data
- Collect genotype data
- QC on genotype data
- Explore data to determine factors affecting molecular trait distribution (covariates, PEERs) and correct for genetic structure (PCs)
- Run association through linear regression with covariates
- Process output of regression model to identify significant associations, correcting for large number of tests performed



Distribution of eQTLs across the genome

- Following association testing perform series of sanity checks on results
 - E.g. map distance of *cis* eQTLs to the TSS expecting clustering of most eQTLs around TSS since region contains a great proportion of regulatory elements

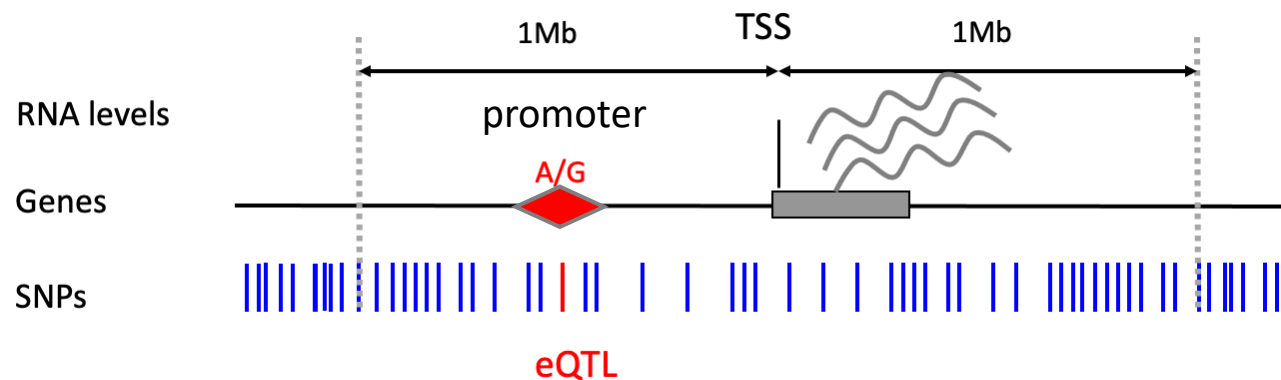


Sanity checks - mapping *cis* eQTLs on genomic features

- Wealth of publicly available information on genomic features in different tissues and cell types provides opportunity to ask: do eQTLs map in known functional annotations?
- Select genomic features from appropriate tissue/cell type and perform enrichment analysis

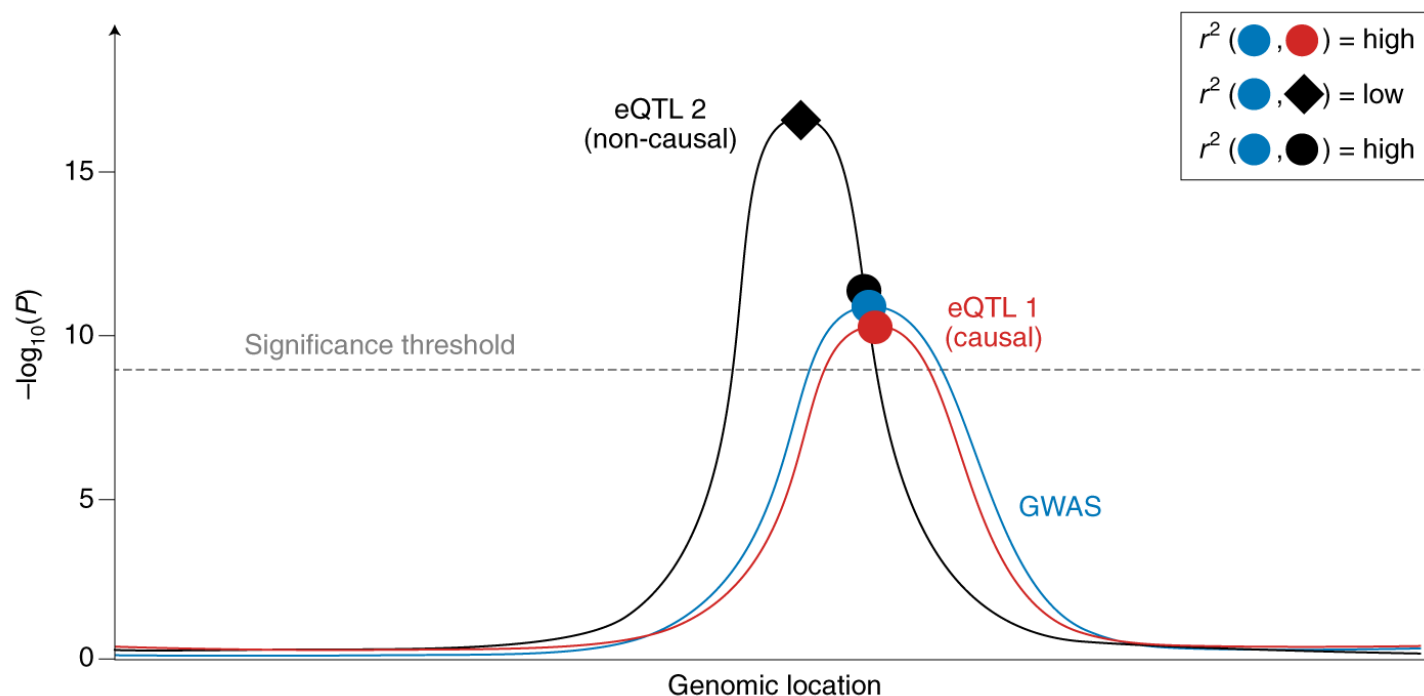
The screenshot displays the 'Regulation' track in the UCSC Genome Browser. It features a grid of genomic annotations, each with a 'hide' button. The annotations include:

- ENCODE Regulation (show button)
- CD34 DnaseI (18)
- CpG Islands
- ENC Chromatin
- ENC DNA Methyl
- ENC DNase/FAIRE
- ENC Histone
- ENC RNA Binding
- ENC TF Binding
- FSU Repli-chip
- GeneHancer
- Genome Segments
- GTEX Combined eQTL
- GTEX Tissue eQTL
- JASPAR Transcription Factors
- NKI Nuc Lamina (18)
- ORegAnno
- Rao 2014 Hi-C
- ReMap ChIP-seq
- Stanf Nucleosome
- SUNY SwitchGear
- SwitchGear TSS (17)
- TFBS Conserved
- TS miRNA Targets
- UCSF Brain Methyl
- UMMS Brain Hist
- UW Repli-seq
- Vista Enhancers

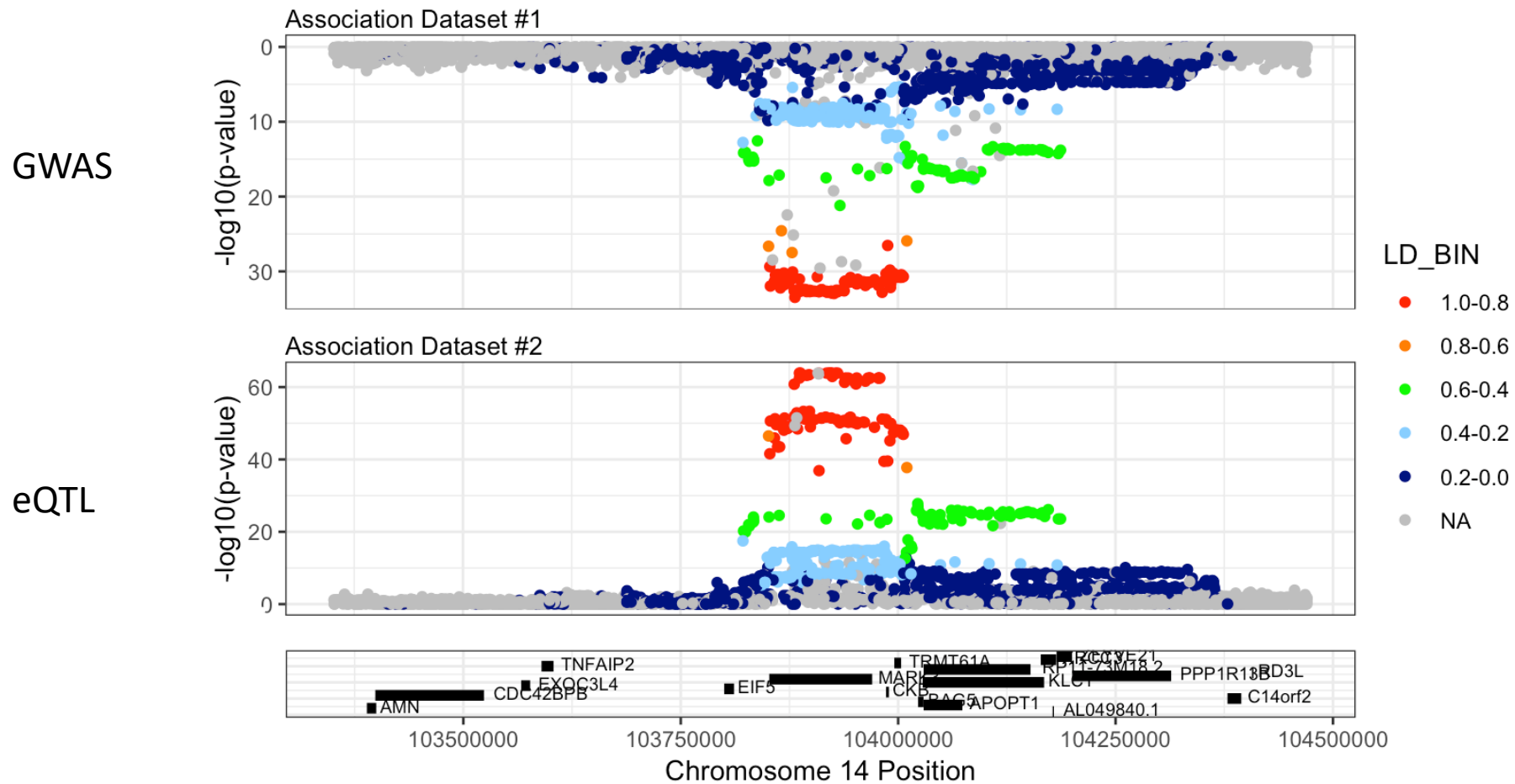


eQTL – GWAS SNP colocalization – is eQTL the causal variant?

- eQTLs identify **variants** affecting **gene expression** / GWAS identifies **variants** affecting **complex traits**
- Signals may overlap in the same region → not always the same causal SNP
- Quantify support for each variant being causal for each trait (GWAS, eQTL)
- Assess whether both signals share the same causal variant (colocalization)



LocusZoom plot example



molQTL resources

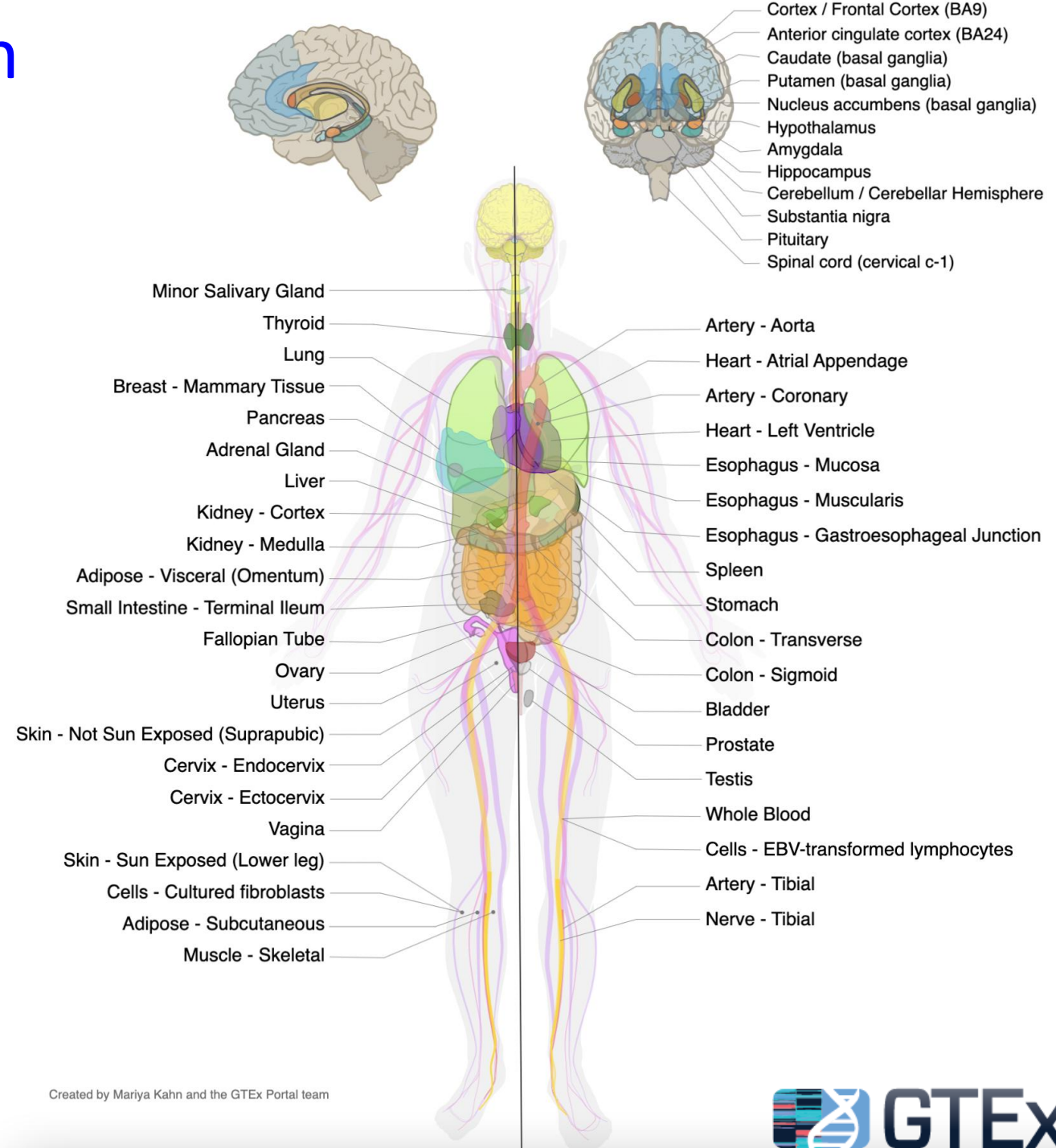
Study/resource	Type	Number of donors	Population ancestries	Biospecimens	Molecular phenotypes
eQTL Catalogue ⁷⁶	Aggregated database of reanalysed data	73–948 per study, total 8,193	88.5% European ancestries	Diverse	Transcriptome phenotypes
GTEX ⁵⁸	Consortium with centralized data production and analysis	73–706	American; 85% European and 11% African ancestries	49 postmortem tissues	Gene expression and splicing, others in smaller scale
eQTLGen ⁵	Consortium with federated analysis	31,684	Predominantly European	Whole blood	Gene expression
GoDMC ⁴²	Consortium with federated analysis	32,851	European ancestries	Whole blood	DNA methylation
Hawe et al. ⁴³	Research project	6,994	European and South Asian ancestries	Whole blood	DNA methylation
Ferkingstad et al. ⁵⁰	Single-cohort study	35,559	Icelandic	Plasma	Aptamer proteomics
Jerber et al. ¹⁵⁸	Research project	215	European ancestries	In vitro differentiated iPSCs	scRNA-seq
Yazar et al. ¹⁵³	Research project	982	European ancestries	PBMCs	scRNA-seq

iPSC, induced pluripotent stem cell; molQTL, molecular quantitative trait locus; PBMC, peripheral blood mononuclear cell; scRNA-seq, single-cell RNA sequencing.

Genotype-Tissue Expression (GTEx) resource

- Public resource to study tissue-specific expression and regulation
 - 54 tissue sites
 - ~1000 individuals
 - WGS, WES, RNA-Seq
- GTEx Portal provides open access to data including gene expression, QTLs, and histology images

How does genetic variation influence gene expression across different human tissues?



INS gene expression levels across 54 tissues

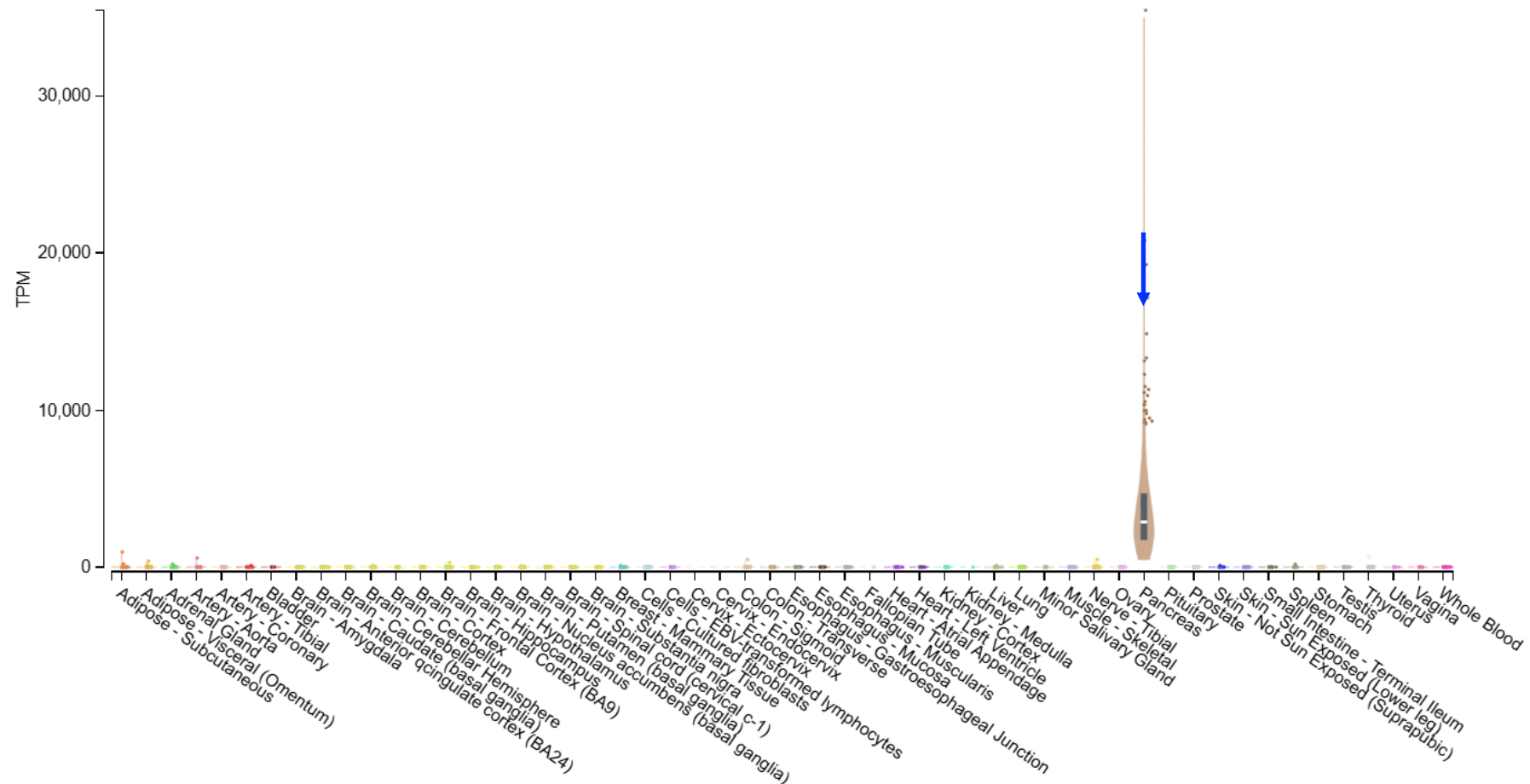
Bulk tissue gene expression for INS (ENSG00000254647.7)

Data Source: GTEx Analysis Release V10 (dbGaP Accession phs000424.v10.p2)

Data processing and normalization ⓘ

Download ▾ SUBSET None Sex SCALE Log Linear TISSUE SORT ▲ ▼ MEDIAN SORT ▲ ▼ OUTLIERS On Off

Bulk tissue gene expression for INS (ENSG00000254647.7)



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